URIIS 2026 - Islamic Economy / Finance & Halal Technology

HALALCHAIN

Blockchain-Enabled Traceability System for Halal Meat Supply Chain Integrity

SCAN. VERIFY. TRUST.

"From label-level assurance to journey-level evidence."

International Islamic University Malaysia

TRL 3 · Functional Lab Prototype

SDG 9 · SDG 12

CAN EVERY HALAL JOURNEY BE VERIFIED?

A label provides assurance. Traceability reveals the journey.

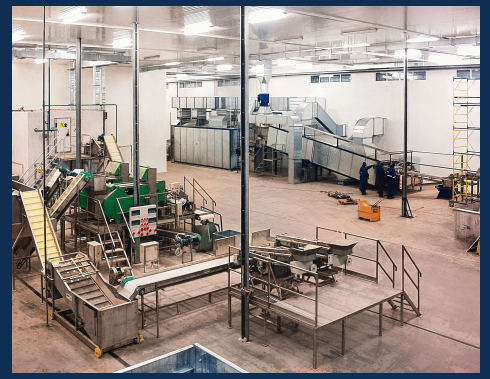


Fragmented information between stages — the physical product moves, but its evidence does not travel with it.

Four Problems on Halal Supply-Chain Trust

Fraud & Adulteration

Mislabelled products, false documentation, substitution or contamination can undermine halal integrity.



Fragmented Records

Supply-chain information may exist across separate exporters, logistics providers, certification records and retailers.



Limited End-to-End Visibility

Stakeholders may see individual stages without having one continuous product journey.



Difficult Verification

Consumers and other verifiers often see the final label rather than the traceability journey behind it.



The challenge is not only certification. The challenge is preserving trustworthy evidence throughout the journey.

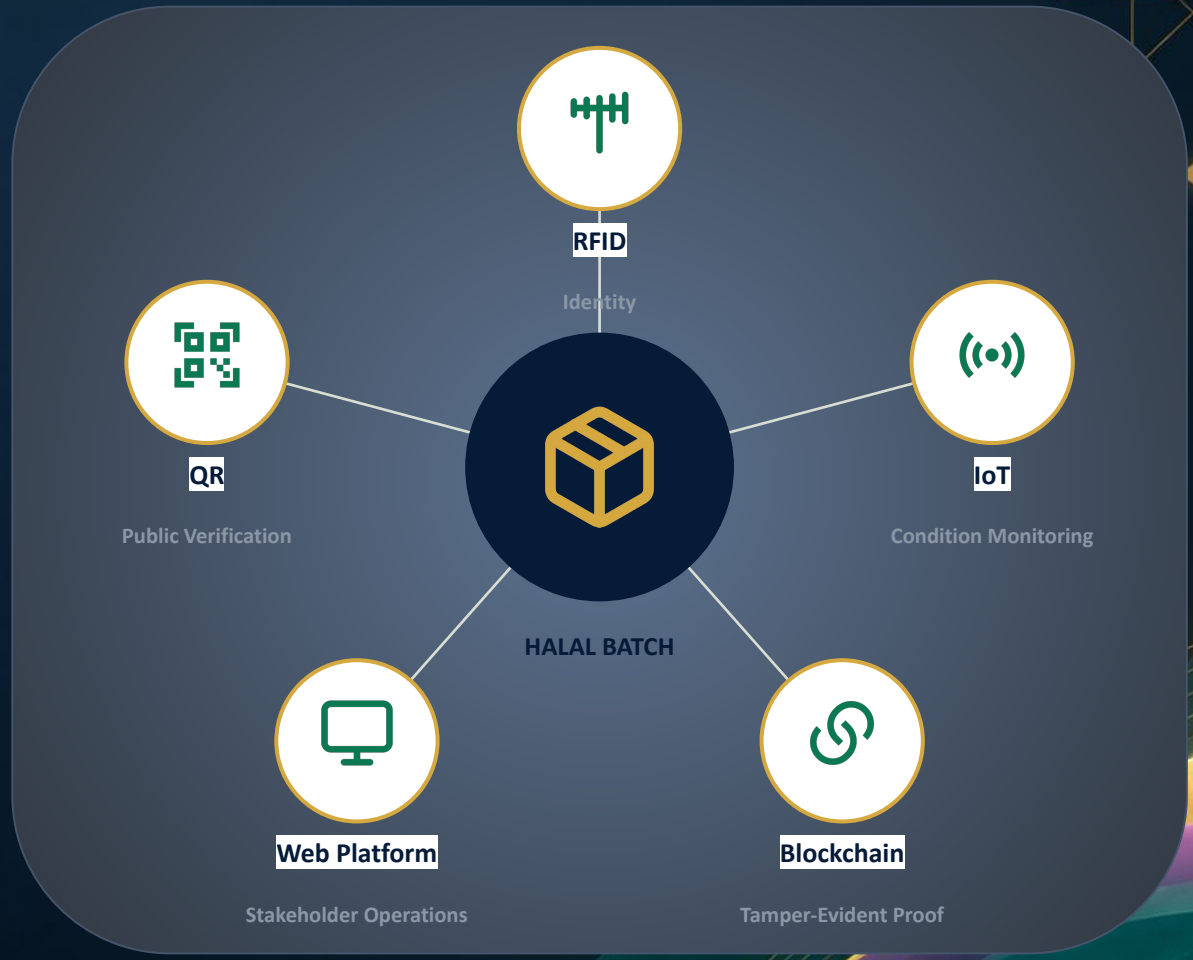
INTRODUCING HALALCHAIN SOLUTION

One Product.
One Digital Identity.
One Verifiable Journey.

HALALCHAIN

A physical-to-digital halal traceability platform connecting physical product movement with digital evidence.

Blockchain is not the entire product — blockchain is the trust layer.



WHAT EXACTLY IS HALALCHAIN ?

Think of Every Halal Batch as Having a Digital Passport.

PHYSICAL HALAL PRODUCT

1. Physical Identity

RFID connects the physical meat batch/container with its digital record.

2. Physical Events

RFID scans and IoT-assisted readings record movements and environmental information.

3. Trust & Proof

Important actions create blockchain-supported records and transaction evidence.

4. Verification

Stakeholder dashboards manage the journey; QR verification exposes a read-only history.

Where Can HALALCHAIN Create Value?

PRIMARY USE CASE: Halal Meat Import / Export Traceability

- Cold-chain visibility
- Shipment checkpoint accountability
- Retail receiving verification
- Halal certificate supporting evidence
- Investigation support
- Audit-support traceability
- Public product verification
- Product recall traceability

FUTURE POTENTIAL: Expansion to Other Halal Products



HOW IT WORKS

Watch the Halal Journey Come Alive

PRODUCER / EXPORTER			LOGISTICS PROVIDER		RETAILER / MFR	PUBLIC VERIFIER
01	02	03	04	05	06	07
Register Batch	Assign QR + RFID	Departure Scan	Checkpoint Scan	Destination Scan	Confirm Delivery	Verify Journey

LIVE STEP · 04 CHECKPOINT SCAN

IN TRANSIT

ACTOR

Logistics Provider

ACTION

RFID tag scanned at checkpoint

DATA CAPTURED

Location, timestamp, sensor reading

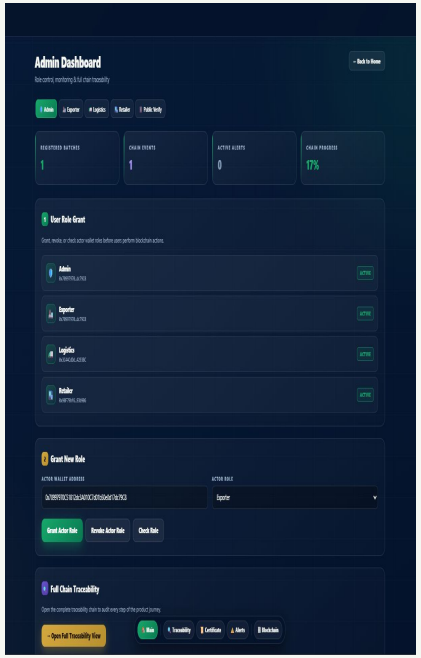
PROOF

Transaction / event evidence recorded

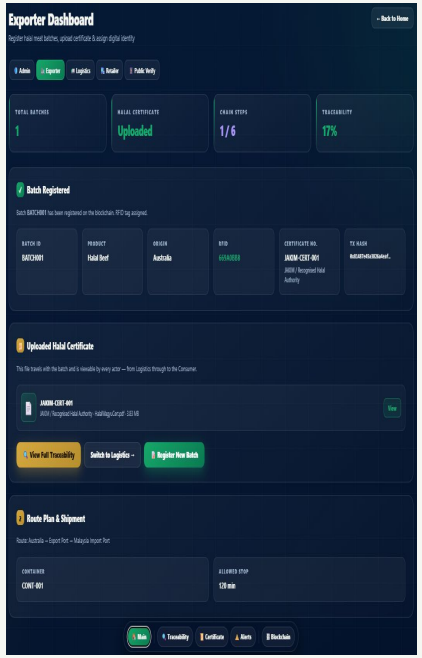
PROTOTYPE IN ACTION

Not Just a Concept. A Functional Lab Prototype.

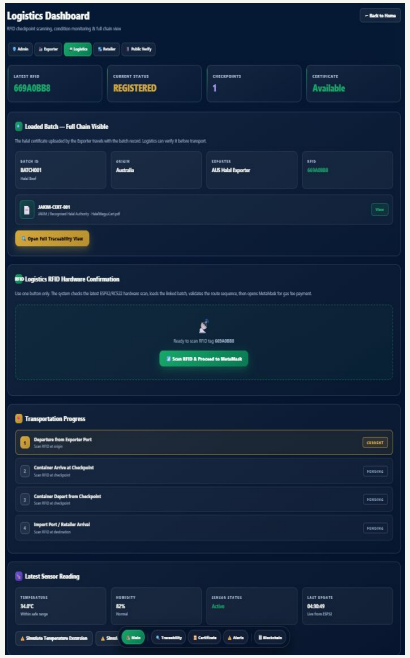
TRL 3



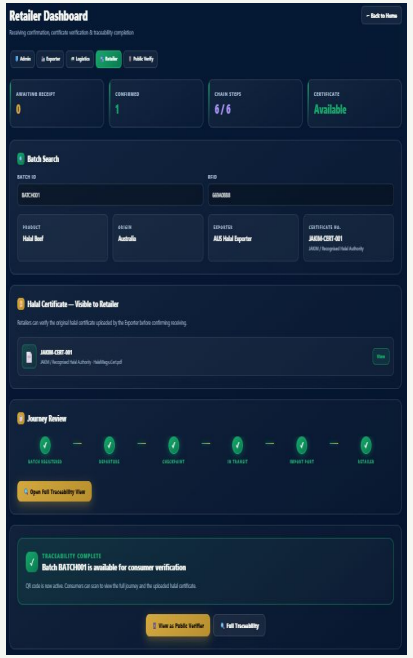
ADMIN



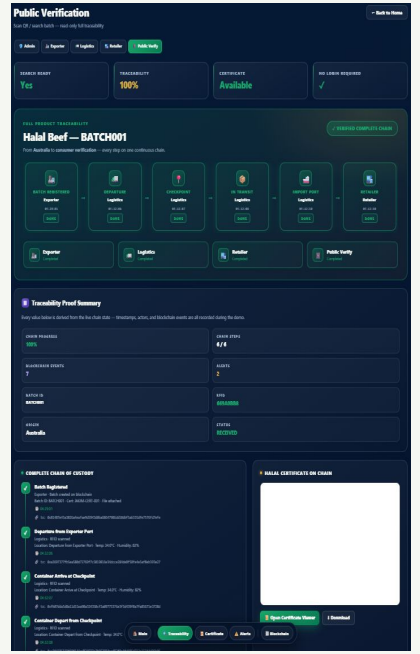
EXPORTER



LOGISTICS



RETAILER



PUBLIC VERIFY

The Right Technology for the Right Problem

PROBLEM	TECHNOLOGY	WHY
Physical Identity	**RFID**	Connects product/container with digital record.
Physical Conditions	**ESP32 + IoT Sensor**	Captures condition information during movement.
Trusted Event Evidence	**Blockchain + Smart Contract**	Creates tamper-evident records and role accountability.
Stakeholder Operations	**React Web Platform**	Provides role-based interfaces.
Public Verification	**QR Code**	Simple read-only traceability access.

CURRENT STACK

FRONTEND

React · Vite · React Router · Axios · Ethers.js · qrcode.react

BACKEND

Node.js · Express · JWT · CORS

BLOCKCHAIN

Solidity · Hardhat · MetaMask · Ethers.js

HARDWARE PROTOTYPE

ESP32 · RC522 RFID · DHT11

From Traceability Data to Trusted Halal Evidence

1. Study problems involving halal meat supply-chain traceability, transparency and immutability.
2. Design an integrated framework using blockchain, RFID and IoT.
3. Develop a functional blockchain-enabled traceability prototype.

TRANSLATING INTO IMPACT

- **More Transparency**
- **Stronger Accountability**
- **Faster Verification**
- **Better Audit Trail**
- **Greater Public Trust**
- **Reduced Opportunity for Undetected Data Manipulation**

STAKEHOLDERS

One Connected Journey. Different Value for Every Stakeholder.

Exporters / Producers

Digital evidence of origin, certificate details, shipment initiation.

Logistics Providers

Checkpoint accountability and movement records.

Importers / Retailers

Review the journey before receiving.

Certification / Audit

Potential supporting digital audit trail.

Government / Import Control

Potential structured traceability evidence.

Consumers

Simple QR-based access to product journey.

VALUE PROPOSITION

The Value Is Not Blockchain.
The Value Is Trust.

TODAY	HALALCHAIN DIRECTION
✕ Fragmented	**Connected**
✕ Manual	**Traceable**
✕ Disconnected	**Evidence-backed**
✕ Reactive	**Accountable**
✕ Harder to verify	**Easier to verify**

HALALCHAIN provides traceability and supporting evidence — it does not itself certify whether meat is halal.

Technology with Responsible Impact

SDG 9

Industry, Innovation and Infrastructure

- Digital infrastructure
- Emerging technology integration
- Halal supply-chain innovation

SDG 12

Responsible Consumption and Production

- Supply-chain transparency
- Accountability
- Accessible consumer information

TRUSTED HALAL ECOSYSTEM

THE VISION

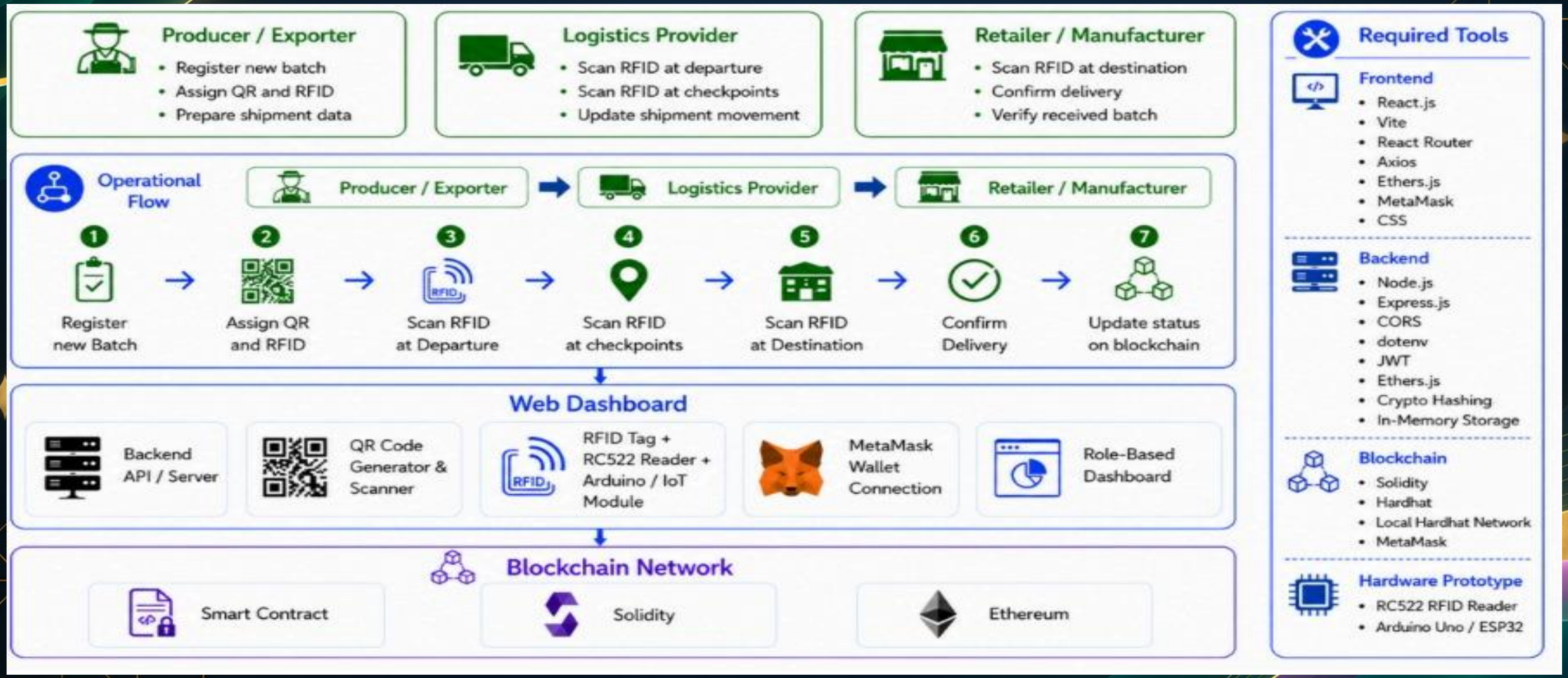
FROM HALAL CERTIFICATION TO CONTINUOUS HALAL TRACEABILITY.



One digital thread connects everyone.

- Real-time sensing
- Persistent data
- Cloud deployment
- Live route information
- Industry-grade cold chain monitoring
- Consortium blockchain evaluation
- Interoperable ecosystem
- Expanded halal product coverage

SYSTEM ARCHITECTURE / FRAMEWORK



READINESS & ROADMAP

Built Now. Ready for the Next Validation Stage.

CURRENT

TRL 3

Functional Lab Prototype

- ✓ Role-based dashboards
- ✓ Blockchain role control
- ✓ Batch registration
- ✓ Certificate information
- ✓ RFID assignment
- ✓ Route plan
- ✓ MetaMask transactions
- ✓ Logistics RFID workflow
- ✓ Checkpoint sequence validation
- ✓ Retailer confirmation
- ✓ Public verification
- ✓ Smart contract events
- ✓ ESP32 + RC522 RFID prototype
- ✓ DHT11 prototype

NEXT

TRL 4–5

Validation Environment

- Persistent database
- Cloud deployment
- Controlled pilot
- Stronger IPFS integration
- Commercial sensor calibration
- GPS connectivity
- Cybersecurity testing
- Industry workflow validation

FUTURE

TRL 6+

Real-World Pilot

- Live halal supply chain
- Multi-company deployment
- Consortium blockchain evaluation
- Enterprise integration
- Broader halal product coverage
- Commercial deployment studies

SWOT ANALYSIS

From Prototype to Industry Platform

STRENGTHS

- Physical-to-digital traceability
- Integrated blockchain + RFID + IoT + QR
- Role accountability
- Public verification
- Functional prototype
- Modular architecture

WEAKNESSES

- Local blockchain
- Prototype storage
- Limited real-world dataset
- Sensor calibration required
- IPFS not fully production-configured
- Stakeholder onboarding required

OPPORTUNITIES

- Halal digitalisation
- Industry pilots
- Government collaboration
- Cold-chain integration
- Broader halal traceability
- Audit-support applications
- International halal trade

THREATS

- Adoption barriers
- Integration challenges
- Infrastructure cost
- Cybersecurity
- Data privacy
- Regulatory requirements
- Competing platforms

COMMERCIALISATION PATH

A Possible Path to Commercialisation

PROPOSED — TO BE VALIDATED WITH INDUSTRY

KEY PARTNERS

- Industry operators
- Logistics companies
- Certification stakeholders
- Technology providers
- Cloud providers
- Universities
- System integrators

KEY ACTIVITIES

- Platform development
- Blockchain
- RFID / IoT integration
- Cybersecurity
- Industry validation

CHANNELS

- Direct industry pilot
- University-industry collaboration
- Halal associations
- Government innovation programmes
- Technology partners

VALUE PROPOSITION

- Connected traceability
- Digital evidence
- RFID identity
- Checkpoint accountability
- Tamper-evident records
- Public verification
- Audit support

CUSTOMER RELATIONSHIPS

- Enterprise onboarding
- Integration
- Training
- Support
- Pilot collaboration

CUSTOMER SEGMENTS

- Exporters
- Importers
- Logistics companies
- Manufacturers
- Retailers
- Certification orgs
- Government stakeholders

KEY RESOURCES

- HALALCHAIN platform
- Smart contracts
- Development team
- Hardware
- Domain expertise
- Industry partners

POSSIBLE REVENUE MODELS (hypotheses) Subscription · Organisation licence · Integration fee · Per-batch service · Hardware package · Maintenance · Analytics modules

THE ASK

We Have the Prototype.
Now We Need the Real-World Environment.

01

Industry Validation Partner

02

Pilot Site

03

Halal Industry Partner

04

Technology Partner

05

Commercialisation Partner

06

Strategic / Government Collaborator

PARTNER WITH US TO TAKE HALALCHAIN FROM TRL 3 TO AN INDUSTRY-VALIDATED PILOT.

SCAN.
VERIFY.
TRUST.

HALALCHAIN

"From source to shelf, make the journey visible."



LET'S BUILD THE NEXT PILOT TOGETHER.